

Solution Requirements

Date	1 July2026
Team ID	xxxxxx
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can directly access the application without login. The system allows anyone to use the crop prediction service.</i>	Low
2	Authorization levels	<i>Single user role. Every user has permission to enter soil and environmental values and receive crop recommendations.</i>	Low
3	External interfaces	<i>The system accepts user inputs through a web interface and interacts with the trained Machine Learning model to generate predictions. No third-party</i>	High

		<i>APIs are required.</i>	
4	Transactions processing	<i>The system collects Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall values, processes them using the trained ML model, and displays the recommended crop.</i>	High
5	Reporting	<i>Displays the predicted crop along with a short description of the recommended crop on the web page.</i>	Medium
6	Business rules, etc.	<i>All seven parameters must contain valid numerical values. Prediction is generated only after successful validation of the inputs.</i>	High
7	Compliance to laws or regulations	<i>The application does not collect or store personal user information and follows standard web application security practices.</i>	Medium
8	Other	<i>Displays meaningful error messages for invalid or missing input values and ensures smooth user interaction.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The application should process user inputs quickly and display predictions without noticeable delay.</i>	<i>Prediction displayed within 2 seconds</i>
2	Scalability	<i>The application should support multiple users simultaneously without affecting performance.</i>	<i>Supports 100+ concurrent users</i>
3	Security & Data Privacy	<i>User input is processed only for prediction and is not permanently stored.</i>	<i>No permanent storage of user data</i>
4	Reliability & Availability	<i>The deployed application should remain available with minimal downtime.</i>	<i>99% availability (depending on hosting platform)</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy for farmers and other users to operate on desktop and mobile devices.</i>	<i>Responsive design with easy-to-understand interface</i>
6	Other	<i>The system should allow easy updates to the machine learning model and application without major code changes.</i>	Modular and maintainable code structure